UNDERSTANDING AND CRAFTING THE MIX

THE ART OF RECORDING

3RD EDITION

WILLIAM MOYLAN

Understanding and Crafting the Mix

Understanding and Crafting the Mix, 3rd edition provides the framework to identify, evaluate, and shape your recordings with clear and systematic methods. Featuring numerous exercises, this third edition allows you to develop critical listening and analytical skills to gain greater control over the quality of your recordings. Sample production sequences and descriptions of the recording engineer's role as composer, conductor, and performer provide you with a clear view of the entire recording process.

Dr. William Moylan takes an inside look at a range of iconic popular music, thus offering insights into making meaningful sound judgments during recording. His unique focus on the aesthetic of recording and mixing will allow you to immediately and artfully apply his expertise while at the mixing desk. A companion website features recorded tracks to use in exercises, reference materials, additional examples of mixes and sound qualities, and mixed tracks.

New to this edition:

- Coverage of basic listening skills and techniques, with integrated, practical exercises
- A chapter on surround sound productions
- Fully updated content reflecting current production concepts and workflows

Dr. William Moylan has worked with leading artists across the full spectrum of jazz, popular, and classical genres. His recordings have been released by major and independent record labels, resulting in wide recognition, including several Grammy award nominations. A leading educator and an active recording engineer and producer for over 30 years, he is a Professor and Coordinator of Sound Recording Technology at the University of Massachusetts Lowell.

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Understanding and Crafting the Mix

The Art of Recording

Third Edition

William Moylan

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Foreword

Nowhere within the mystery of creation is the concept of infinity more closely demonstrated than in the human response to sound.

Sounds barely audible in the quiet solitude of a forest glade contain information about direction, height, distance, and character, that unconsciously provide awareness of our surroundings.

Sounds as you step into a great cathedral, hearing nearby soft footfalls on ancient flagstones and the singing of a distant choir, that provide clues of distance and perspective, and even invite sharing of mood.

Sounds in a concert hall, an office, a bathroom, or a recording studio, each with its own message prompting our response.

Blindfold, the smallest sample provides amazing awareness of our environment.

The way sound behaves within a space paints a picture. We don't have to analyze, measure, or evaluate. Created in the image of a communicating God, we communicate with speech to express what we think and with music to express what we feel.

Music is as old as man. References to music, song, poetry, and musical instruments go back thousands of years. We see Egyptian *bas-reliefs* and know that music played an enormous part in their culture and religion. Near the end of King David's reign, the Hebrews had professional temple choirs and a 4,000-strong orchestra.[1] We read of strolling minstrels and court musicians of later ages. But what did they sound like? Every age of man is recorded in writing, painting, and sculpture, but there are no sound recordings.

Today we can choose from vast libraries of music from any part of the world, which have become the benchmarks of artistic and technical quality for many thousands of people who may never have even been to a concert. Technical quality has improved enormously over the years but, surprisingly, we can still enjoy the very earliest recordings with all their imperfections of noise, distortion, limited bandwidth, and poor dynamic range. Why, then, is *The Art of Recording* important?

As technology advances, realism has become more accurate, but it seems that the artistic qualities of a performance have become more elusive. Neither accuracy nor artistry can be tabulated in some book of rules. Unlike hardware design—for example, computers, where precision and speed predominate and the figures accurately define the performance—in the field of sound recording, specifications and measurements cannot describe the sounds that we hear and neither can they predict the effect of those sounds on us. The recordist must provide the vital link as our interpreter.

In the 1953 fourth edition of his *Radio Designer's Handbook*, Langford-Smith states: "It is common practice to regard the ear as the final judge of fidelity, but this can only give a true judgment when the listener has acute hearing, a keen ear for distortion and is not in the habit of listening to distorted music. A listener with a keen ear for distortion can only cultivate this faculty by making frequent direct comparisons with the original music in the concert hall." [2]

We must cultivate "A Point of Reference."

Truly successful recordists and producers have developed very high degrees of refinement and can perceive qualities (or the lack of them!) in the sound recording and reproducing chain that seem to defy reason and sometimes to contradict our current state of knowledge.

In 1977, Geoff Emmerick, who with George Martin recorded The Beatles at Abbey Road and later at Air Studios in London, showed me that he could hear a difference between two identical channels on a recently delivered new console. After some hours of listening with him, I agreed that I could hear a subtle difference. When we measured, I found that out of 48 channels, 3 had been incorrectly terminated and displayed a rise of 3 dB at 54 kHz. The limit of hearing for most humans does not extend beyond 20 kHz, and this small resonance, whilst obviously an oversight in the factory, would not normally have been regarded as important.

One of the significant features of this episode was that Geoff was deeply "unhappy," even "distressed" at what he was hearing or perceiving. Since then I have seen much more evidence that the range beyond 20 kHz is part of human awareness. Newly introduced designs that transmit frequencies to beyond 100 kHz (with low distortion and noise) surprisingly sound warmer, sweeter, and fuller.

In 1987, when addressing the Institute of Broadcast Sound in London, I carried out a simple experiment for the first time, with the object of discovering what effect frequencies above 20 kHz might have on a professionally aware audience.

A generator capable of switching between a sine and a square wave was fed through an "ordinary" amplifier to an "ordinary" monitor loudspeaker. The frequency was set at 3 kHz. The audience confirmed that they could hear the third harmonic as a superimposed 9 kHz tone or "whistle" when the generator was switched to square wave. (A square wave contains predominantly odd harmonics. When the first of these exceeds the limit of hearing, the sine and square wave should sound the same.)

The frequency was then progressively raised and older members soon admitted that they could no longer hear the third harmonic "whistle." But all could still hear a *difference* in quality as the generator was switched between sine and square. As the frequency continued to be slowly raised, some could still identify a difference when the fundamental had reached 15 kHz.

This experiment has been repeated many times in different parts of the world without any real attempt at scientific control. "Ordinary" equipment provided by my host was used on every occasion. Results have been surprisingly consistent: some 35 to 45 percent of those present were able to identify a quality difference when the fundamental was as high as 15 kHz.

There was one exception: At the University of Massachusetts Lowell, some 60 percent of the audience were still identifying a difference when the fundamental frequency had exceeded 17 kHz.

At the 83rd Convention of the Audio Engineering Society, Dr. William Moylan proposed a "Systematic Method for the Aural Analysis of Sound Sources."[3] Dr. Moylan uses his method at the University of Massachusetts Lowell Department of Music. I think this points to the success of Dr. Moylan's training in aural analysis! This same method has led to this book.

No measurements or formulae can ever replace the recordist, but he must develop a reliable point of reference and learn "the Art of Recording." The learning process is endless. We never arrive at an ultimate state of knowledge. We are, even now, only scratching the surface, and this is especially so as we explore new formats to convey not only technical excellence but a whole listening experience of music and its environment.

Can there be a perfect recording?

Only if we can arrive at perfect knowledge. How, then, should such knowledge be used? Could it change, for example, the world in which we live?

Stephen Hawking, in *A Brief History of Time*,[4] seeks a unified theory—drawing together the general theory of relativity and of quantum mechanics—which would lead to a complete understanding of the events around us and of our own existence. He says: "If we find the answer to that, it would be the ultimate triumph of human reason—for then we would know the mind of God." We need open minds to *envision* direction, responsible minds to *choose* direction, and a point of reference beyond ourselves to whom we are ultimately answerable.

Well over one hundred years ago, Lord Rayleigh told us that the ears are the final arbiter of sound: "Directly or indirectly, all questions connected with this subject must come for decision to the ear, as the organ of hearing; and from it there can be no appeal. But we are not, therefore, to infer that all acoustical investigations are conducted with the unassisted ear. When once we have discovered the physical phenomena which constitute the foundation of sound, our explorations are, in great measure, transferred to another field lying within the dominion of the principles of Mechanics. Important laws are in this way arrived at, to which the sensations of the ear cannot but conform."[5]

To follow William Moylan's "important laws" will prepare your ears and your mind for the true "Art of Recording." His approach is proven and will lead the reader's ears to the refined levels of "keen-ness" and "acuteness" required today. Dr. Moylan's book also gives us insight into what turns "recording" into art, and provides ways to bring artistic sensibility into our work.

Rupert Neve
Wimberley, TX
September 2001

Endnotes

1 *Bible*. 1 Chronicles 23:5. " . . . and four thousand are to praise the LORD with the musical instruments I have provided for that purpose."
2 F. Langford-Smith, editor, *Radio Designer's Handbook,* 4th edition, Sydney, New South Wales: Wireless Press for Amalgamated Wireless Valve Company Pty. Ltd, 1953, Ch.14, Section 12, (iii).
3 William Moylan, "A Systematic Method for the Aural Analysis of Sound Sources in Audio Reproduction/Reinforcement, Communications and Musical Contexts," paper read at the 83rd Convention of the Audio Engineering Society, October, 1987.
4 Stephen W. Hawking, *A Brief History of Time* (Toronto: Bantam Press, 1992), pp. 173–175.
5 Lord Rayleigh, The Theory of Sound, 1st edition, 1877, New York: Dover Publications, 1945.

Preface

This third edition is expanded and largely reworked from the second edition. Perhaps only 20% of the second edition remains untouched here.

This edition improves the functioning of the previous editions with simplified language and more detailed coverage of some subjects and layout of materials. It adds new and important features, and seeks to be more useful and pertinent to the intended audience: you.

These are the greatest differences. First, the book has been expanded to four parts, with a new Part Two on background fundamental listening skills. A new chapter dedicated to surround sound brings extensive coverage of the sound stage dimensions of surround, and examines the musical impacts of various approaches to surround. The number of exercises has expanded, more thoroughly covering both developing listening skills and becoming aware of the creative potentials of production. These latter exercises support the greatly expanded coverage of the artistic elements in crafting the mix.

A companion website has been established for this edition, and replaces the CD of the second edition. It can be found at www.focalpress.com/cw/moylan. This website will allow additional and more detailed evaluations of recordings and graphics for the reader. It will supplement the text in supplying templates for certain graphs and exercises. There will be more tracks available on the website, as well as the original tracks from the CD. It is expected that other sound files will be added over time to refresh the content without revising the text in the short term.

While the purpose and goals of *Understanding and Crafting the Mix: The Art of Recording* have not changed, I believe this writing is now significantly more thorough and effective, clearer and more readable.

Background

Understanding and Crafting the Mix: The Art of Recording is the product of my experiences as an educator in sound recording technology (music and technology), and of my thought processes and observations as a composer of acoustic music and of music for recordings (recording productions and electronic music). It includes what I have learned through my creative work as a record producer and through my attempts to be a facile and transparent recording engineer. It is informed by my perspective as a music theorist and a teacher of music analysis. This book is an outgrowth of my on-going and in-depth research into how we hear music and sound as reproduced through loudspeakers (aural perception, music cognition) and aural skills pedagogy, and into the analysis of recording related to the dimensions of recorded music and their impact on the music, its materials, expression, and meaning.

This book has evolved substantially since my initial research and writings in the early 1980s. It has been greatly shaped by years of devising instructional methods and materials in my courses at the University of Massachusetts Lowell, and by my interaction with many other audio educators and observations of other recording programs worldwide. This evolution has been enhanced by many other people and experiences as well; I have been fortunate to have interacted with numerous talented production professionals in the audio industry, representing many different types and sizes of facilities (in both major and idyllically isolated locations) and many production styles—with very diverse and deeply impressive credits and credentials.

The concepts and methods of this book have gone through many stages of development over the past three decades. They will continue to be refined as we learn more about listening and what we hear, how we perceive sound and understand art, and how we learn.

Purpose and Intended Audience

Understanding and Crafting the Mix: The Art of Recording seeks to bring the reader to understand how recorded sound is different from live sound, and how those differences can enhance music. It will bring the reader to explore how those sound characteristics appear in significant recordings. The book also presents a system for developing critical and analytical listening skills necessary to recognize and understand these sound characteristics.

This leads to the production process itself. *Understanding and Crafting the Mix* seeks to move the reader to engage audio recording as a creative process. Techniques and technologies are purposely not covered. Instead, the book explores the recording process as an act of creating art, and

helps the reader envision recording devices as musical instruments. It seeks to develop an artistic sensitivity that will lead the reader to find and create their own unique artistic voice in shaping and creating music recordings—neutral to the continual shift of technologies and techniques, devices and software, fashion and convention.

Unchanged from the first edition, *Understanding and Crafting the Mix: The Art of Recording* is intended to be used: (1) as a resource book for all people involved in audio; (2) as a textbook for courses in recording analysis, critical listening (listening skills-related courses), audio-production-related courses of all types in sound recording technology, music engineering, music technology, media/communications, or related programs; or (3) as a self-learning text for the motivated student, beginning professional, or interested amateur. It can also provide a unique look at some of the most important recordings by The Beatles, for all who may be interested. It has been written to be accessible to people with limited backgrounds in acoustics, engineering, physics, math, and music.

The intended reader might be an active professional in any one of the many areas of the audio recording industry, or a student studying for a career in the industry. The reader might also be learning recording through self-directed study, or be anyone interested in learning about recording and recorded sound, perhaps an audiophile.

The portions of the book addressing sound-quality evaluation will be directly applicable to all individuals who work with sound. Audio engineers in technical areas will benefit from this knowledge, and the related listening skills, as much as individuals in creative, production positions. Even those involved in consumer audio and pro audio sales can benefit. All those people who talk about sound can make use of the approach to evaluating sound that is presented.

As a resource book, *Understanding and Crafting the Mix* is designed to contribute to the professional development of recordists. The book seeks to clearly define the sonic dimensions of audio recording, and brings the reader to approach recording production creatively. It offers to expand the current professionals' creative thinking, their skills in critical and analytical listening, and their skill in and sensitivity to accurate and meaningful communication about sound quality.

Many excellent books exist on recording techniques and audio technologies. Articles in many excellent magazines, journals, and serials exist that cover recording devices and techniques, audio technologies, and acoustic concerns. These areas are not addressed here.

Little has been written on the aesthetic and creative aspects of recording music or audio production. Few books exist that discuss and develop

the listening skills required to evaluate recordings for technical quality, and none to evaluate the artistry of recordings or to analyze the artistic content of recordings. This book seeks to fill this void. It may be used as a sourcebook or a textbook in all of these areas, from beginning through to the most advanced levels.

The book is well suited to developing the student's music-production skills and artistry. It is designed to stimulate thought about the recording process as being a collection of creative resources, and to invite a creative spirit. These skills and creative ideas can then be applied to the act of crafting the music recording artistically.

Graphing the activity of the various artistic elements is important for developing listening and evaluation skills, especially during beginning studies. It is also valuable for performing in-depth evaluations of recordings that allow us to study how accomplished record producers have used the artistic elements to achieve their "sounds." This process of graphing the activity of the various artistic elements is also a useful documentation tool—functional even with the less exacting detail that would make it more practical for daily use. Working professionals through to beginning students will find something useful in the approach.

Finally, it is hoped this book will clarify communication in some small way and in some small segment of our industry.

All audio professionals communicate "about" sound in some way, for some reason. Still, we often do not communicate well, clearly, or accurately. Perhaps this book might serve as a meaningful point of departure for an "audio recording syntax" to be devised.

Selection of Recorded Examples: The Beatles

Over the years I have received many well-intentioned comments that my writings would benefit from more current musical and recorded examples than my almost exclusive use of Beatles records. Most comments have been very articulate, and made great sense. I have always seen much to support this idea, on its surface. Bringing this into practice is something else. I have struggled with this almost daily in my teaching, as I wish my examples to be of relevance to my younger audience.

Current music has many additional social and musical dimensions that make this difficult to navigate. I find when I bring current recordings to listening sessions, people have difficulty separating the artist (and what the artist represents to those individuals) from the sounds they hear (they turn off their ears, or distort their perception). Further, preference for certain types of music can run deep with young people; deeper, still,

are their dislikes. Then the question becomes: What artists, or types of music, or recording styles should I use? Should it be Taylor Swift, Eminem, Will-i-am, U2, Dixie Chicks, James Taylor, Linkin Park, Red Hot Chili Peppers, Neil Diamond, or [place *your* favorite artist here]? Which of these would you like to listen to, perhaps 30 times, to fully appreciate the supplied graphs?

I venture to estimate that any recording selection may alienate 75% of this readership for one or more personal reasons.

The first book I published that discussed many of this book's topics was released in 1991. Musical examples included tracks from the then current and now classic albums *The Joshua Tree*, *So*, *Face Value*, *Every Breath You Take*, as well as important songs by Elton John, Tracy Chapman, and Kate Bush. It was interesting (and disconcerting) to recognize: Only three years after its publication the book *looked* out of date because of the musical examples, though its content was not. The recordings that brought it to continue to appear most relevant were those that were not fixed in the time of publication: those of The Beatles. This gave me much to consider.

Now I keep returning to The Beatles, for a great many reasons that are deeper than surface. First, as teaching tools and sonic examples, they continue to work perfectly well in demonstrating fundamental concepts and sound dimensions that are permanent in the field. As early stereo and multitrack recordings, many sound qualities and dimensions of sound are clearer, and more easily perceived by novice listeners, than those found in many later recordings.

The matter of preferences to listen to certain types of music can be mostly circumvented with The Beatles. The Beatles are simply part of "our" common experience and common language—across generations, across time periods. While rooted in the 1960s, they reach far beyond—and well before. Just as the music of Coltrane or Bach, they are timeless in their expression, though a product of their time. They simply "are," in much the same way as the music of Beethoven just is—whether you like it or do not like it is irrelevant, these recordings are important and should be known. They are touchstones of music, and of the recording industry. Everyone knows them, and can talk intelligently about them—if you don't, you are at a disadvantage in this field.

Of course, new production techniques are not found in The Beatles' records. Exceptions are the surround mixes of *LOVE* reflecting more current production practice and the reissue of The Beatles' catalog remastered in 2009 with its more contemporary timbral balances. This writing, however, is not "about" presenting the reader with examples of modern recordings; it is about educating the ears and providing

knowledge to understand "any" recording—most importantly, to prepare the reader to engage those recordings and styles that have yet to be created, especially their own.

And, the songs, recordings, and performances really are well constructed on a great many levels. Stylistically diverse, they offer a great many vantage points into production sound and techniques, as well as music. The public's curiosity has brought many reissues, each unique in some way. All of this allows us to study the topics of this book with some consistency of material—material that can simply be accepted as a given, that is well crafted, and that was masterfully recorded. I do not believe these statements to be subjective or personal opinion.

On a personal level, I do not consider myself an unbridled "fan" of The Beatles, blinded by their mystique and fanatically obsessed with the Fab Four. This is often assumed of me.

Rather, I am immensely indebted to their work, and what it has taught me and brought me. And not only the four composer/performers: Sir George Martin, Geoff Emerick, and all the other production staff of all the releases educated my ears and my awareness of the impossible world of records. I learned a significant portion of what I know as a recordist by studying the production techniques and sounds of their records. Beginning with experiencing their appearance on *The Ed Sullivan Show* live at age eight and listening to their records in my formative years, through struggling to figure out their unusual sounds and to play their newly released "Strawberry Fields Forever" at age eleven, to poring over their recording processes as a fledgling assistant engineer in my early twenties, their music has helped shape my musicianship, my career, my artistic sensibilities, and my sense of self—and more. The intersections of this music, of these records, and my career continue to this day. I hold a deep sense of gratitude for all they have provided me.

Acknowledgments

It has been nearly 14 years since I joined the list of Focal Press authors, and I have been blessed to have worked with many fine editors. My profound thanks to all of my editors during these years, most of whom have moved on: Terri Jadick, Marie Lee, Tricia Tyler, Catharine Steers, Emma Baxter, Beth Howard, Diane Wurzel, and of course Meagan White and Megan Ball. My deepest appreciation to the other wonderful editorial staff and production people at Focal Press—who might be too numerous to mention but who have all made my work easier and more enjoyable by their professionalism and constant willingness to help. I am honored to be affiliated with this extraordinary organization and group of people.

For over 30 years now, my students in the Sound Recording Technology program at the University of Massachusetts Lowell have worked through the materials and concepts of this book in any number of forms, and have taught me a great deal. I extend a special thanks to these serious and gifted young people for their (largely unknowing) contributions to my work.

Many other people have provided support for this book, or assisted in shaping my thoughts about these materials. While they are far too many to mention, they are all very important to me. Only a very few follow.

My gratitude to Paul Angelli, Patrick Drumm, Erh-Chaun Lai, Gavin Paddock, Phil Reese, Brad Swanson, Michael Testa, and Mark Whittaker, who gave much time and extended considerable effort working through specific graphs and exercises.

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Introduction

The recording process shapes music. Recording techniques and technologies change the acoustic sounds they capture, and impart new sound characteristics. These sound qualities are under the control of an individual who shapes the music recording—the recordist.

The sound qualities created by recording do not occur in nature, though some qualities simulate natural dimensions or relationships. These qualities are sonic dimensions that are unique to audio recordings, and give recorded music (or music reproduced over loudspeakers) a set of characteristic sound characteristics. While these dimensions create an experience that is different from a live, unamplified performance, these sound qualities have become accepted as part of the experience of listening to recorded and reproduced sound, and of that music itself. Considering that vastly more recorded or amplified music is heard today than live, entirely acoustic music, it is readily understood that these sound qualities of recorded sound are now integral parts of today's music—and of the listening experience of today and of recent decades.

These sound qualities have become part of the fabric of recorded music. The way a piece of music might be shaped has thus been extended to include the unique sound qualities of music recordings.

The person controlling, or creating, these sound qualities (the recordist) is therefore functioning as a creative artist. This person is a musician of the technology and process—"conducting" by encouraging and ensuring quality performances, "performing" recording, mixing, and processing devices, and "composing" the mix's sound qualities and relationships.

As this book unfolds, the answers to the following questions will gradually become evident, as we begin *understanding the mix*, consider *crafting the mix*, and navigate the dimensions of *the art of recording*:

- What makes recording music an art?
- What makes the music recording a unique medium for artistic expression?
- What is different between a music recording and a live, acoustic-music performance?
- What has brought the recordist recognition as an artist?
- How does the recording process contribute to the sound of a piece of music?

Overview of Organization and Materials

This book will define the unique aspects of sound that are captured, created, and shaped in the recording process, and will examine their appearance in well-known recordings by The Beatles. A system for developing listening skills and understanding the artistry of music recordings will then be presented, unfolding throughout the middle chapters of the book. Through this process and its many related exercises the reader will be brought to gain understanding of artistic aspects of audio recordings, and control over shaping them in the recording process. Ultimately, the reader will learn to better understand and utilize these qualities to establish their own artistic voice as a recordist.

Accordingly, the book is divided into four parts:

- Part One defines the artistic dimensions and sound characteristics of music recordings.
- Part Two builds fundamental listening skills towards hearing these dimensions of recorded sound.
- Part Three leads to an understanding of how those characteristics are commonly used in production practice, and carries the reader through a system for listening-skill development to recognize nuance in those dimensions.
- Part Four explores the artistry of recording processes, and leads the reader conceptually through the journey and the artistry of crafting a music recording.

Part One. Defining the Art of Recording: The Sound Characteristics and the Aesthetic Qualities of Audio Recordings

Part One is divided into three chapters. To begin defining the artistic dimensions of music recordings, sound itself will be explored. The states

of sound in air, in human perception, and as applied to music are followed in the sequence of understanding the meaning of sounds. The processes of moving from one state of sound to another are covered. The anomalies that occur in these transfer processes are recognized and evaluated.

Sound as a resource for artistic expression is the basis for Chapter 2. This chapter explains the unique sound qualities of recordings, and the potential of those qualities to be used in artistic expression.

This is followed by an examination of the musical message itself, leading to how musical materials are perceived by the listener, and how that perception leads to the understanding of musical messages and to communication. In many ways this foreshadows the last chapter of the book.

Part Two. Learning to Listen, Beginning to Hear: Acquiring Fundamental Listening Skills and Establishing an Effective Approach to Listening

Part Two is also divided into three chapters. The focus of this part is to bring the reader to learn the skill of listening, so they may begin to consistently hear the dimensions presented in Part One. Chapter 4 discusses the listening process, its importance and role, personal development, and "talking about sound."

Chapter 5 leads the reader through a good number of fundamental listening skills. This is a core set of skills and experiences that will bring the reader/listener the background needed to engage the more complex tasks that follow. As a whole, acquiring these skills will require some significant time and attention. The chapter's eleven exercises are carefully crafted to effectively guide the reader through acquiring these skills. The skills are all important, as they prepare the reader for Part Three—and their career.

The book's system for evaluating sound is outlined in Chapter 6. A typical sequence for exploring materials, found to be most functional, is offered. Graphing dimensions of sound against time is central to this analytical approach and system. Creating a timeline and recognizing sound sources against time and structure presented here will be used throughout Part Three.

The exercises presented in this chapter can be undertaken in any order. Some readers will find some exercises much easier than others; mastery of one is not required before moving on. The reader is strongly encouraged to continue returning to exercises that pose a challenge, until the material is learned and the reader is comfortable that the skill is controlled.

Part Three. Understanding the Mix: Developing Listening and Sound-Evaluation Skills

The six chapters of Part Three present the complete system for evaluating the dimensions of sound in music and audio recordings that was outlined in Part Two.

Each element of sound is evaluated separately, with stereo and surround separated into two chapters. A method of evaluation has been specifically devised for each individual element, and reflects how the elements appear in music recordings. Commercially available and widely celebrated recordings by The Beatles provide the examples of how the unique sound qualities of recordings have enhanced the music and have at times contributed in fundamental ways to shaping musical ideas.

The methods of evaluation for each individual sound quality are accomplished in relation to a complete, interrelated system of evaluating sound in music recordings. A series of listening exercises is presented throughout the course of these chapters to guide the reader in developing sound-evaluation and listening skills. This culminates by examining the evaluations of all elements, at their various levels of detail, for their interrelationships.

Starting in Part Two and continuing throughout Part Three, the reader progresses from simple concepts and listening processes to the most complex. The sequencing builds on experiences that are likely the most easily learned and evolves systematically to those typically most challenging. Listening experiences that many audio professionals or intermediate-level musicians may have already acquired (at least intuitively) are incorporated.

As the system develops the reader's listening skills, it will establish a basis for meaningful and accurate communication on sound content and quality. This is a significant outcome from learning this system.

An objective vocabulary and a way to evaluate sound have been devised; this allows precise information about sound to be recognized and communicated. This information will reflect the actual states or values of the sound material. Subjective impressions and subjective terms about the sound's quality (a very common way musicians and recordists attempt to communicate about sound) are always avoided; they do not allow communication to be accurate and severely limit its value. The method avoids any personal impressions about the qualities of sound by addressing only the physical dimensions of sound as they are perceived and as they appear in the music. The method for evaluating sound will allow individuals to talk "about" sound in meaningful ways. It will allow people to exchange precise and conclusive information about the sound qualities, once they acquire the skills required to recognize those qualities.

The diligent reader will gradually gain the experience and the knowledge that will allow them to perform quite complex listening and evaluation tasks, and to describe sounds to others.

Time, practice, understanding, repetition, and concentrated effort are all required to develop listening skills. Auditory memory will increase as the listener becomes more accustomed to the sound material, more aware of patterns of levels and changes within all of the artistic elements, and more skilled at focusing attention on the aspects of sound we are conditioned from birth to ignore. Developing refined listening skills is a long-term project; the individual's listening skills will likely continue to develop throughout their career.

Part Four. Crafting the Mix: Shaping Music and Sound, and Controlling the Recording Process

Part Four brings the reader to apply these learned sound qualities to crafting the mix. How these sound qualities contribute to mixes is explored, including how they interrelate in practice. These five chapters bring the reader to make use of the production process to shape their own recordings with artistic sensitivity.

Part Four begins by examining the roles of the recordist, both functional and artistic. The first chapter continues with the production process shaping the recording aesthetic, its relationship to the live listening experience, and the altered realities of performance created by the production process.

Next we examine constructing and crafting the mix by bringing attention to individual artistic elements, and groupings of elements, as musical materials. The contributions of elements are examined for how they shape and impact the music and its expression. These are the concepts and materials engaged in composing the mix.

The intersections of practical, technical considerations with artistic decisions are encountered by examining microphones, equipment selection and technologies, monitoring and selecting timbres within Chapter 15. How these play out in practice is mirrored in the next chapter's coverage of production sequencing: Every project is unique, and requires unique choices of sound and methodology; some stray further from typical choices than others. Tracking and recording; editing, compilations, and signal processing; and preparing for the mix take us through preparation and planning the mix.

Part Four concludes the text with "Crafting the Mix, and Finalizing the Production." Here mix considerations meet musical materials and expression, and the story of the song. The mix is conceptualized to support the form and structure of the music and the text. Mastering and

the listener's experience finish the chapter and the book. Many readers might have purchased this book specifically for this chapter; should the reader wish, this last chapter could be read first—though reading in the presented sequence is recommended, and the reader would do well to return to the first chapter after reading the last first.

It is a goal of *Understanding and Crafting the Mix: The Art of Recording* to bring the reader to explore the creative potentials of the medium's tools (equipment) as musical instruments. Exercises are contained throughout Part Four to engage the reader in learning the subtleties of performing the mix, and crafting its sound qualities. The concepts are encountered singly and build, thus allowing for their exploration for artistry as well as function.

Companion Website

A companion website has been constructed to support this book at www.focalpress.com/cw/moylan. On it is a core set of tracks to illustrate many key concepts of the book. The reader will be referred to the website for these tracks throughout the text. Text boxes, such as the one below, are inserted throughout the book to identify specific tracks to hear and the purpose for listening.

The uncompressed audio files on the website are the intended versions for use. It is imperative the reader download these uncompressed files for use in listening exercises. Compressed files are present only to allow the reader the option of auditioning a track before downloading the larger, uncompressed file. The compressed files will alter or eliminate much of the material that is being discussed, and are unusable for study and understanding the material. The compressed files could make for advanced listening exercises, though, in learning to recognize the differences created by compression by comparing them to the uncompressed files.

The many recorded examples on the website will assist in developing important listening skills, and some tracks will present dimensions of recorded sound in ways that are more easily recognized than in many commercial recordings. The reader will often want to listen to tracks while reading specific sections in the text; they are meant to clarify materials and make some of the more abstract concepts much more tangible.

The website also contains additional materials, including graphs and descriptions of recordings other than those explained in the text. Materials will continue to be added over time to refresh the website, as well as to enhance the book's content and the reader's experience.

Three tracks are included on the website to aid in establishing an accurate playback system. Readers are strongly encouraged to carefully evaluate their sound systems. Accurate and quality playback of the website tracks,

and of the commercial recordings cited throughout the book, is necessary if the reader is to understand the materials being presented.

Establishing an Accurate Playback of Recordings

A reasonable-quality, accurate playback system located in an acceptable listening environment is needed to experience and learn the material discussed in this book. It is important that the recorded examples be heard as intended—with a minimum of change from the playback system and the room—in order to learn the materials being discussed. The reader should have access to such a system in order to benefit from the readings. Headphones are not a suitable substitute for accurate listening to nearly all of the examples.

High-resolution monitors in an acoustically neutral environment and an impeccable signal chain are required for many activities performed by professional recordists. It is not at all realistic to expect a beginner in the industry, a student, or an interested amateur to have such a system. It is, however, necessary for the reader to hear the website tracks and the commercial recordings cited throughout the book with some semblance of accuracy.

Putting together a quality sound system and establishing an accurate playback environment are covered in Chapter 15. Topics presented are:

- Components and specifications for a high-quality playback system to provide unaltered, detailed sound.
- Loudspeaker placement and listening-room interaction.
- Listener distance and orientation to loudspeakers.
- Monitoring levels.

If readers are uncertain about having a suitable playback system, they are encouraged to look over that material now, or certainly before playing any of the listening examples.

The sound-system evaluation tracks (tracks 54–56) on the website can help the reader evaluate their current system. It is strongly suggested that readers do so in order to know the performance of their system, and to ensure it is properly set up. It is also recommended they acquire a sound-level meter. The meter will not only assist in evaluating their system, but will also help the reader to work through a number of exercises in this book and will lead the reader to establish accurate and healthy monitoring practices.

Listen

to tracks 54–56

for playback system set-up and calibration material; read the track descriptions for assistance.

If a reader cannot establish a reasonable quality of playback on their own, they are strongly encouraged to find a suitable location where they can listen to the recordings that are cited. This is an important part of learning the materials presented.

Summary

Part One provides the necessary background to recognize the qualities of sound. It defines the sound characteristics and aesthetic qualities of audio and music recordings. It brings the reader to understand that the recording process enhances music recordings, and has the potential to shape their artistic qualities substantially.

Parts Two and Three provide a framework and a method to develop high-level listening skills. It will lead the listener to develop listening and sound-evaluation skills and to learn how to communicate objective and meaningful information about sound. Important commercial recordings are examined to illustrate the sound characteristics being studied.

Part Four brings focus to the art and aesthetics of recording, provides some context to understand the recordist's place within any project, and guides the creative processes of conceiving, composing, performing, and crafting the mix. While this section will not present prescriptions for production techniques to achieve certain results, or provide information on selecting and using specific equipment or technologies, it will lead the reader to directed discovery and channeled creativity. It will bring the reader to gain control of the craft of recording music and shaping sound, and to use the recording process in an artistic way—no matter the technology, techniques, music genre: those of yesterday, today, or tomorrow.

Understanding and Crafting the Mix: The Art of Recording encourages the reader to think broadly and deeply about the creative, artistic, and technical processes, and it provides the guidance to enable them to do so. The reader will be brought to conceive the music recording as a self-contained piece of music, rather than a product of technological processes.

The goal of *Understanding and Crafting the Mix: The Art of Recording* is to bring the reader to an awareness of the dimensions of recorded sound, and to an understanding of the dimensions as they appear in music recordings, through the development of listening and evaluation skills. With these acquired, it seeks to lead readers to find, establish, and refine their own unique artistic voice in shaping and crafting music recordings.

Part One

Defining the Art of Recording:

The Sound Characteristics and the Aesthetic Qualities of Audio Recordings

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1 The Elements of Sound and Audio Recording

Audio recording is the recording of sound. It is the act of capturing the physical dimensions of sound and then reproducing those dimensions. The dimensions are reproduced either immediately or from a storage medium (magnetic, vinyl, electronic, digital), and thereby returning those dimensions to their physical, acoustic state.

Audio recording is a process that transforms sound from its physical acoustic state, through the recording/reproduction chain, and back to physical sound.

The "art" of recording emerges with the artistically sensitive application of the recording process. The recording process is being used to shape or create sound as an artistic statement (piece of music), or supporting artistic material. To be in control of crafting the artistic product, one must be in control of the recording process, be fluent in the ways the recording process modifies sound, and be skilled in executing well-defined creative ideas.

These all closely involve interaction with sound. Inconsistencies between the various states of sound are present throughout the audio-recording process. Many of these inconsistencies are the result of the human factor: the ways in which humans perceive sound and interpret or formulate its meanings. In order for the artist (audio engineer/recordist) to be in control of the artistic processes they must understand the substance of their material: sound, in all its inconsistencies.

The States of Sound

In audio recording, sound is encountered in three different states. Each of these three states directly influences the recording process. These three states are:

1. Sound as it exists physically (having physical dimensions);
2. Sound as it exists in human perception (psychoacoustic conception): sound perceived after being transformed by the ear and interpreted by the mind (the perceived parameters of sound being human perceptions of the physical dimensions); and
3. Sound as idea: sound as an abstract or a tangible concept, as an emotion or feeling, or representing a physical object or activity (this is how the mind finds meaning from its attention to the perceived parameters of sound); sounds as meaningful events, capable of communication, provide a medium for artistic expression; sounds hereby communicate, have meaning.

The audio-recording process ends with sound reproduced over loudspeakers, as sound existing in its physical state, in air. Often the audio-recording process will begin with sound in this physical state, to be captured by a microphone.

The *recordist*—the person making the recording—and all others involved in the industry listen to the sound in all facets of the audio-recording process. They evaluate the audio signal at all stages while the recording is being made. This, of course, results in the recording that is heard by the end listener, and is the reason for making the recording. In the listening process we translate the physical dimensions of sound into the perceived parameters of sound through the listening process (aural perception).

This translation process involves the hearing mechanism functioning on the physical dimensions of sound, and the transmission of neural signals to the brain. The process is nonlinear and alters the information; the hearing mechanism does not produce nerve impulses that are exact replicas of the applied acoustic energy.

Certain aspects of the distortion caused by the translation process are, in general, consistent between listeners and between hearings; they are related to the physical workings of the inner ear or the transfer of the perceived sounds to the mind/brain. Other aspects are not consistent between listeners and between hearings; they relate to the listener's unique hearing characteristics and their experience and intelligence.

The final function occurs at the brain. At a certain area of the cortex, the neural information is processed, identified, consciously perceived, and stored in short-term memory; the neural signals are transferred to

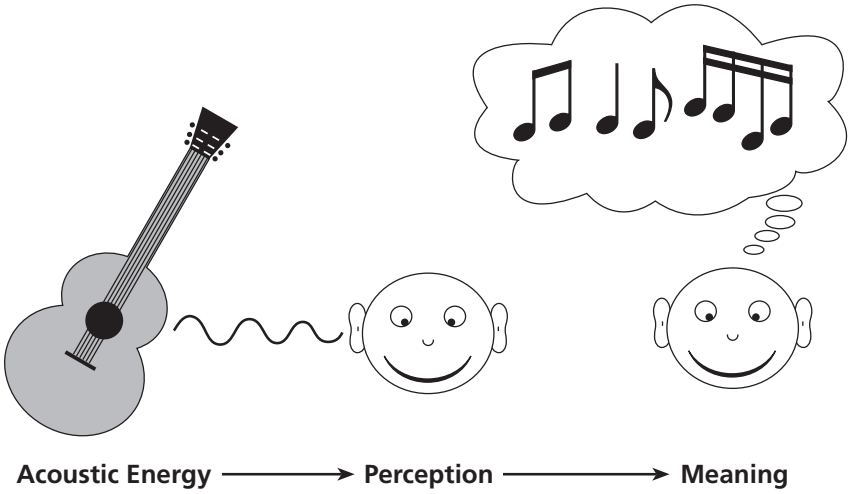


Figure 1.1
Three states of sound: in air, in perception, as message.

other centers of the brain for long-term memory. At this point, the knowledge, experience, attentiveness, and intelligence of the listener become factors in the understanding and perception of sound's artistic elements (or the meanings or message of the sound). The individual is not always sensitive or attentive to the material or to the listening activity, and the individual is not always able to match the sound to their previous experiences or known circumstances.

Sequentially, the physical dimensions (1) are interpreted as perceived parameters of the sound (2). The perceived parameters of sound (2) provide a resource of elements that allow for the communication and understanding of the meaning of sound (and artistic expression) (3).

The audio-recording process communicates ideas, and can express feelings and emotions. Audio might take the form of music, dialog, motion-picture action sounds, whale songs, or others. Whatever its form, audio is sound that has some type of meaning to the listener. The perceived sound provides a medium of variables that are recognizable and have meaning when presented in certain orders or patterns. Sound, as perceived and understood, becomes the resource for creative and artistic expression. The artist uses the perceived parameters of sound as the artistic elements of sound, to create and ensure the communication of meaningful (musical) messages.

In the next section the individual states of sound as physical dimensions and as perceived parameters will be discussed individually. The interaction of the perceived parameters of sound will follow the discussion of the individual parameters. These discussions provide critical information for understanding the breadth of the "artistic elements of sound" in audio recording, presented in the next chapter.

Physical Dimensions of Sound

Five physical dimensions of sound are central to the audio-recording process. These physical dimensions are: the characteristics of the sound waveform as (1) *frequency* and (2) *amplitude* displacements, occurring within the continuum of (3) *time*; the fusion of the many frequency and amplitude anomalies of the single sound to create a global, complex waveform as (4) *timbre*; and the interaction of the sound source (timbre) and the environment in which it exists creates alterations to the waveform according to variables of (5) *space*.

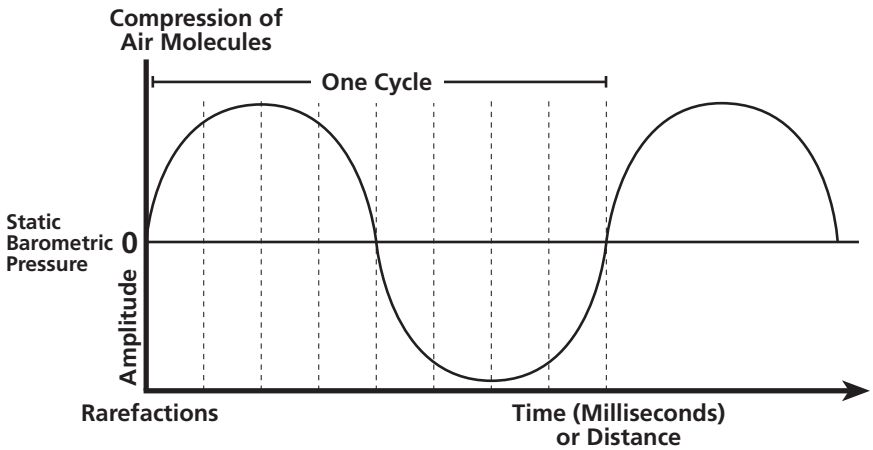


Figure 1.2
Dimensions of the waveform.

Frequency is the number of similar, cyclical displacements in the medium, air, per time unit (measured in cycles of the waveform per second, or Hz). Each similar compression/rarefaction combination creates a single cycle of the waveform. Amplitude is the amount of displacement of the medium at any moment, within each cycle of the waveform (measured as the magnitude of displacement in relation to a reference level, or decibels).

Timbre

Timbre is a composite of a multitude of functions of frequency and amplitude displacements; it is the global result of all the amplitude and frequency components that create the individual sound. Timbre is the overall quality of a sound. Its primary component parts are the dynamic envelope, spectrum, and spectral envelope.

The *dynamic envelope* of a sound is the contour of the changes in the overall dynamic level of the sound throughout its existence. Dynamic envelopes of individual acoustic instruments and voices vary greatly in

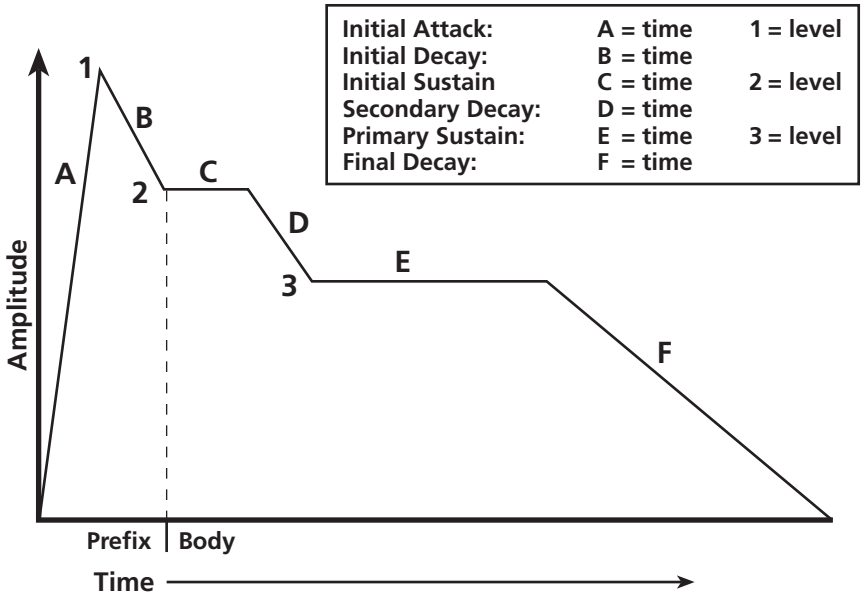


Figure 1.3
Dynamic envelope.

content and contours. The dynamic envelope is often thought of as being divided into a number of component parts. These component parts may or may not be present in any individual sound. The widely accepted components of the dynamic envelope are: attack (time), initial decay (time), initial sustain level, secondary decay (time), primary sustain level, and final decay (release time).

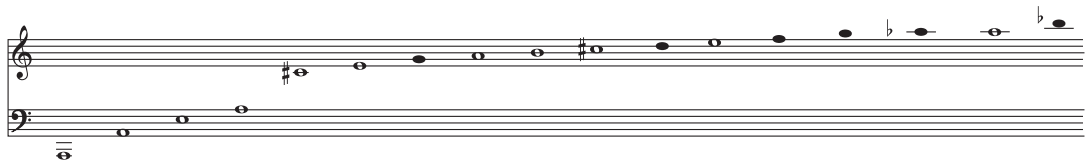
Dynamic envelope shapes other than those created by the above outline are common. Many musical instruments have more parts to their characteristic dynamic envelope, and many instruments have fewer. Further, vocalists and the performers of many instruments have great control over the sustaining portions of the envelope, providing internal dynamic changes to sounds. Musical sounds that do not have some variation of level during the sustain portion of the envelope are rare; the organ is one such exception.

The *spectrum* of a sound is the composite of all of the frequency components of the sound. It is comprised of the fundamental frequency, harmonics, and overtones, sometimes including subharmonics and subtones.

The periodic vibration of the waveform produces the sensation of a dominant frequency. The number of periodic vibrations, or cycles of the waveform, that repeat its characteristic shape is the *fundamental frequency*. The fundamental frequency is also that frequency at which the sounding body resonates along its entire length. The fundamental

frequency is often the most prominent frequency in the spectrum, and will often have the greatest amplitude of any component of the spectrum.

In all sounds except the pure sine wave, frequencies other than the fundamental are present in the spectrum. These frequencies are usually higher than the fundamental frequency. They may or may not be in a whole-number relationship to the fundamental. Frequency components of the spectrum that are whole-number multiples of the fundamental are *harmonics*; these frequencies reinforce the prominence of the fundamental frequency (and the pitched quality of the sound). Those components of the spectrum that are not proportionally related to the fundamental we will refer to as *overtones*. Traditional musical acoustics studies define overtones as being proportional to the fundamental, but with a different sequence than harmonics (first overtone = second harmonic, etc.); this traditional definition is herein replaced by a differentiation between *partials* that are proportional to the fundamental (harmonics) and those that are not (overtones). This distinction will prove important in the evaluation of timbre and sound quality in later chapters. All of the individual components of the spectrum are partials. Partials (overtones and harmonics) can exist below the fundamental frequency as well as above; they are accordingly referred to as subharmonics and subtones.



Harmonic =	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Frequency =	55Hz	110	165	220	275	330	385	440	495	550Hz	605	660	715	770	825	880	935 Hz
Pitch =	A_1	A_2	E_3	A_3	$C^{\sharp}_4$	E_4	G_4 (flat)	A_4	B_4	$C^{\sharp}_5$	D_5 (sharp)	E_5	F_5 (sharp)	G_5 (flat)	$A^{\flat}_5$ (flat)	A_5	$B^{\flat}_5$ (sharp)

Figure 1.4
Harmonic series.

For each individual instrument or voice, certain ranges of frequencies within the spectrum will be emphasized consistently, no matter the fundamental frequency. Instruments and voices will have resonances that will strengthen those spectral components that fall within these definable frequency ranges. These areas are called *formants*, *formant regions*, or *resonance peaks*. Formants remain largely constant, and modify the same frequency areas no matter the fundamental frequency. Spectral modifications will be present in all occurrences of the sound source with harmonics or overtones in the formant regions. Formants can appear as increases in the amplitudes of partials that appear in certain frequency bands, or as spectral components in themselves (such as noise transients caused by a hammer striking a string). They can also

be associated with resonances of the particular mechanism that produced the source sound. Formants are largely responsible for shaping the characteristic sounds of specific instruments; they allow us to differentiate between the instruments of different manufacturers, or even to tell the difference between two instruments of the same model and maker.

Listen

to tracks 1 and 2

for the harmonic series played in individual frequencies and pitches, and as a chord.

A sound's spectrum is composed primarily of partials that create a characteristic pattern. This pattern is recognizable as being characteristic of a particular instrument or voice. This pattern of spectrum will transpose (change level but maintain the same distances between frequencies/partials) with every new fundamental frequency of the same sound source and remain mostly unchanged. In this way, this consistent pattern will form a similar timbre at different pitch levels. Formants establish frequency areas that will be emphasized for a particular instrument or voice. These areas will not change with varied fundamental frequency, as they are fixed characteristics (such as resonant frequencies) of the device that created the sound. Formants may also take the form of spectral information that is present in all sounds produced by the instrument or voice.

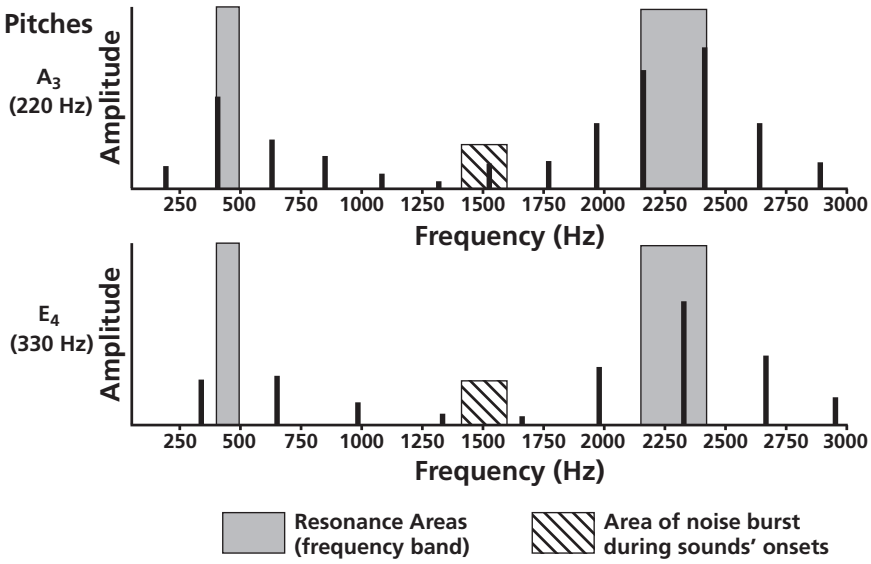


Figure 1.5 Format regions of two pitches from a hypothetical instrument. Vertical lines represent the partials of the two pitches, placed at specific frequencies and at specific amplitudes.

The frequencies that comprise the spectrum (fundamental frequency, harmonics, overtones, subharmonics, and subtones) all have different amplitudes that change independently over the sound's duration. Thus, each partial has a different dynamic envelope. Altogether these dynamic envelopes of all the partials make up the *spectral envelope*. The spectral envelope is the composite of each individual dynamic level and dynamic envelope of all of the components (partials) of the spectrum.

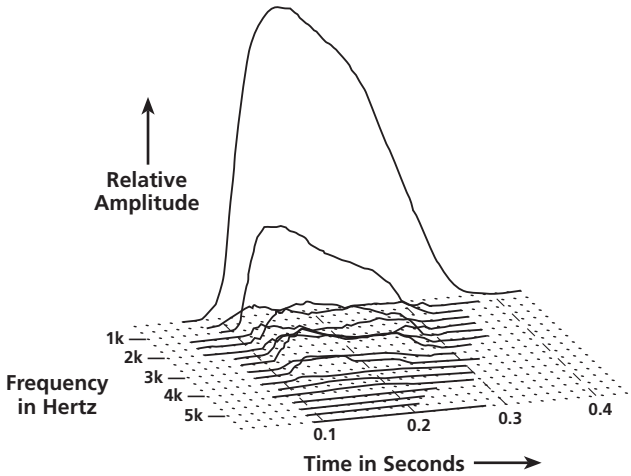


Figure 1.6
Spectral envelope.

The component parts of timbre (dynamic envelope, spectrum, and spectral envelope) display strikingly different characteristics during different parts of the duration of the sound. The duration of a sound is commonly divided into two time units: the *prefix* or *onset*, and the *body*. The initial portion of the sound is the prefix or onset; it is markedly different from the remainder of the sound, the body. The time length of the prefix is usually determined by the way a sound is initiated, and is often the same time unit as the initial attack. The actual time increment of the prefix may be anywhere from a few microseconds to 20–30 ms.

The prefix is defined as the initial portion of the sound that has markedly different characteristics of dynamic envelope, spectrum, and spectral envelope than the remainder of the sound. The body of the sound is usually much longer in duration than the prefix. (See Figure 1.3.)

Space

Sounds are changed by the environments in which they are produced. The interaction of the sound source (timbre) and the environment will create alterations to sound. These changes to the sound source's sound quality are created by the acoustic space. The nature of these alterations

is directly related to (1) the characteristics of the acoustic space in which the sound is produced and (2) the location of the sound source within the environment.

Space-related sound measurements must be performed at a specific physical location. The measurements are calculated from the point in space where a receptor (perhaps a microphone or a listener) will capture the composite sound (the sound source within the acoustic space). The location of the listener (or other receptor) becomes a reference in the measurement of the acoustic properties of space.

The aspects of space that influence sound in audio recording are: (1) the *distance* of the sound source to the listener, (2) the *angle* of the sound source to the listener, (3) the *geometry of the environment* in which the sound source is sounding, and (4) the *location* of the sound source *within the host environment*.

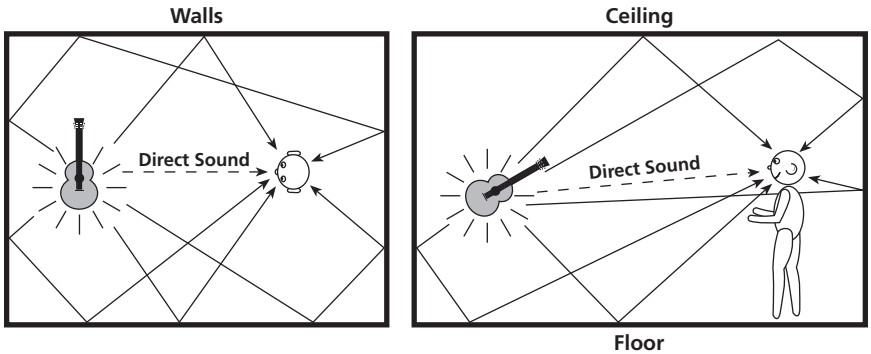


Figure 1.7
Paths of reflected sound within an enclosed space.

The environment in which the sound source is sounding is often referred to as the *host environment*. Within the host environment, sound will travel on a direct path to the listener (as *direct sound*) and sound will bounce off reflective surfaces before arriving at the listener (as *reflected sound*).

Reverberant sound is a composite of many reflections of the sound arriving at the listener (or microphone) in close succession. The many reflections that comprise the reverberant sound are spaced so closely that the individual reflections fuse into a dense set of reflections and create a single sonic impression. These many reflections are therefore considered as a single entity. As time progresses, these closely spaced reflections become even more closely spaced and of diminishing amplitude, until they are no longer of consequence. *Reverberation time* (often referred to as RT60) is the length of time required for the reflections to reach an amplitude level of 60 dB lower than that of the original sound source.